PROBLEM SET 8

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Problem to demonstrate the fitting of a binary response variable using logistic regression

a. Consider the Weekly data available in the ISLR package of R. Use the data from 1990 to 2008 as train data and the data on 2009-2010 as test data.

```
rm(list=ls())
library(ISLR)
library(caret)
dir_bin=ifelse(Weekly$Direction=="Up",1,0)
wk_data=cbind(Weekly[,-9],dir_bin)
attach(wk_data)
head(wk_data)
```

```
##   Year   Lag1   Lag2   Lag3   Lag4   Lag5    Volume  Today dir_bin
## 1 1990  0.816  1.572 -3.936 -0.229 -3.484 0.1549760 -0.270       0
## 2 1990 -0.270  0.816  1.572 -3.936 -0.229 0.1485740 -2.576       0
## 3 1990 -2.576 -0.270  0.816  1.572 -3.936 0.1598375  3.514       1
## 4 1990  3.514 -2.576 -0.270  0.816  1.572 0.1616300  0.712       1
## 5 1990  0.712  3.514 -2.576 -0.270  0.816 0.1537280  1.178       1
## 6 1990  1.178  0.712  3.514 -2.576 -0.270 0.1544440 -1.372       0
```

```
str(wk_data)
```

```
## 'data.frame':    1089 obs. of  9 variables:
##  $ Year   : num  1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 ...
##  $ Lag1   : num  0.816 -0.27 -2.576 3.514 0.712 ...
##  $ Lag2   : num  1.572 0.816 -0.27 -2.576 3.514 ...
##  $ Lag3   : num  -3.936 1.572 0.816 -0.27 -2.576 ...
##  $ Lag4   : num  -0.229 -3.936 1.572 0.816 -0.27 ...
##  $ Lag5   : num  -3.484 -0.229 -3.936 1.572 0.816 ...
##  $ Volume : num  0.155 0.149 0.16 0.162 0.154 ...
##  $ Today  : num  -0.27 -2.576 3.514 0.712 1.178 ...
##  $ dir_bin: num  0 0 1 1 1 0 1 1 1 0 ...
```

```
train_df=wk_data[Year<2009,]
test_df=wk_data[Year>=2009,]
```

b. Using the training data, fit a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Are there any significant predictors? If not, which of the tests is reporting the least p value. Interpret the coefficient corresponding to that test.

```
model_full=glm(dir_bin~Lag1+Lag2+Lag3+Lag4+Lag5+Volume,data=train_df,family=b
inomial(link="logit"))
summary(model_full)
```

```
##
## Call:
## glm(formula = dir_bin ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
##     family = binomial(link = "logit"), data = train_df)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.33258    0.09421   3.530 0.000415 ***
## Lag1        -0.06231    0.02935  -2.123 0.033762 *
## Lag2         0.04468    0.02982   1.499 0.134002
## Lag3        -0.01546    0.02948  -0.524 0.599933
## Lag4        -0.03111    0.02924  -1.064 0.287241
## Lag5        -0.03775    0.02924  -1.291 0.196774
## Volume      -0.08972    0.05410  -1.658 0.097240 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1354.7  on 984  degrees of freedom
## Residual deviance: 1342.3  on 978  degrees of freedom
## AIC: 1356.3
##
## Number of Fisher Scoring iterations: 4
```

The logistic regression results indicate that none of the predictors are statistically significant at the 5% level, as all p-values exceed 0.05. However, Lag2 has the smallest p-value, suggesting it is the most influential predictor in the model. Its positive coefficient implies that an increase in Lag2 leads to an increase in the log-odds, and hence the probability, of the market moving "Up", indicating a positive association with the response.

c. Estimate the odds of market going "up" at median values of the predictors.

```
med_vals=apply(train_df[,2:7],2,median);med_vals

##     Lag1     Lag2     Lag3     Lag4     Lag5   Volume
## 0.231000 0.234000 0.231000 0.230000 0.230000 0.804848

med_vals=data.frame(t(apply(train_df[,2:7],2,median)));med_vals

##    Lag1  Lag2  Lag3 Lag4 Lag5   Volume
## 1 0.231 0.234 0.231 0.23 0.23 0.804848

logit_val=predict(model_full,newdata=med_vals)
odds_val=exp(logit_val);odds_val

##        1
## 1.267475

table(test_df$dir_bin)#actual count of the directions
```

```
##
##  0  1
## 43 61
```

e. Using the fitted model and the predictor values from the test data, find an optimal cut point that minimizes TPR(1 − FPR). Report the confusion matrix corresponding to the optimal cut-point value. From the confusion matrix compute the test error rate.

```
prob_full=predict(model_full,newdata=test_df[,2:7],type="response")
tpr_full=c()
fpr_full=c()
metric_full=c()
for(i in 1:length(prob_full)){
  pred_full=ifelse(prob_full<prob_full[i],0,1)
  cm_full=confusionMatrix(as.factor(pred_full),as.factor(test_df$dir_bin))
  tpr_full[i]=cm_full$byClass["Sensitivity"]
  fpr_full[i]=1-cm_full$byClass["Specificity"]
  metric_full[i]=tpr_full[i]*(1-fpr_full[i])}
out_full=data.frame("Cut_Point"=prob_full,"TPR"=tpr_full,"FPR"=fpr_full,"Comp
are"=metric_full)
head(out_full)

##     Cut_Point       TPR       FPR    Compare
## 986 0.3900305 0.1162791 0.1639344 0.09721693
## 987 0.6084798 0.9534884 0.9836066 0.01563096
## 988 0.4489399 0.4186047 0.3770492 0.26077011
## 989 0.4101072 0.2325581 0.2131148 0.18299657
## 990 0.4336596 0.3488372 0.2786885 0.25162028
## 991 0.4406039 0.3488372 0.3442623 0.22874571

out_full[which.max(out_full$Compare),]

##      Cut_Point       TPR      FPR   Compare
## 1043 0.4579394 0.5116279 0.442623 0.2851697

opt_cut_full=out_full[which.max(out_full$Compare),1]
cat("The cutpoint for which TPR(1-FPR) is minimum:",opt_cut_full)

## The cutpoint for which TPR(1-FPR) is minimum: 0.4579394

pred_opt_full=ifelse(prob_full<opt_cut_full,0,1)
cm_opt_full=confusionMatrix(as.factor(pred_opt_full),as.factor(test_df$dir_bi
n))
cm_opt_full

## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##          0 22 27
##          1 21 34
##
```

```
##                Accuracy : 0.5385
##                  95% CI : (0.438, 0.6367)
##     No Information Rate : 0.5865
##     P-Value [Acc > NIR] : 0.8631
##
##                   Kappa : 0.0676
##
##  Mcnemar's Test P-Value : 0.4705
##
##             Sensitivity : 0.5116
##             Specificity : 0.5574
##          Pos Pred Value : 0.4490
##          Neg Pred Value : 0.6182
##              Prevalence : 0.4135
##          Detection Rate : 0.2115
##    Detection Prevalence : 0.4712
##       Balanced Accuracy : 0.5345
##
##        'Positive' Class : 0
##
```

f. Is there any change in the test error rate if we have worked with only the two predictors lag 1 and lag 2?

```
model_red=glm(dir_bin~Lag1+Lag2,data=train_df,family=binomial)
prob_red=predict(model_red,newdata=test_df,type="response")
tpr_red=c()
fpr_red=c()
metric_red=c()
for(i in 1:length(prob_red)){
  pred_red=ifelse(prob_red<prob_red[i],0,1)
  cm_red=confusionMatrix(as.factor(pred_red),as.factor(test_df$dir_bin))
  tpr_red[i]=cm_red$byClass["Sensitivity"]
  fpr_red[i]=1-cm_red$byClass["Specificity"]
  metric_red[i]=tpr_red[i]*(1-fpr_red[i])}
out_red=data.frame("Cut_Point"=prob_red,"TPR"=tpr_red,"FPR"=fpr_red,"Compare"
=metric_red)
head(out_red)

##     Cut_Point        TPR        FPR    Compare
## 986 0.4386143 0.04651163 0.03278689 0.04498666
## 987 0.6934260 0.97674419 1.00000000 0.00000000
## 988 0.5539158 0.51162791 0.42622951 0.29355700
## 989 0.5209042 0.27906977 0.21311475 0.21959588
## 990 0.5338470 0.46511628 0.26229508 0.34311857
## 991 0.4728669 0.06976744 0.06557377 0.06519253

out_red[which.max(out_red$Compare),]

##      Cut_Point       TPR       FPR   Compare
## 1078 0.5275291 0.4418605 0.2131148 0.3476935
```

```r
opt_cut_red=out_red[which.max(out_red$Compare),1]
cat("The cutpoint for which TPR(1-FPR) is minimum:",opt_cut_red)
```

```
## The cutpoint for which TPR(1-FPR) is minimum: 0.5275291
```

```r
pred_opt_red=ifelse(prob_red<opt_cut_red,0,1)
cm_opt_red=confusionMatrix(as.factor(pred_opt_red),as.factor(test_df$dir_bin)
)
cm_opt_red
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction  0  1
##          0 19 13
##          1 24 48
##
##                Accuracy : 0.6442
##                  95% CI : (0.5443, 0.7357)
##     No Information Rate : 0.5865
##     P-Value [Acc > NIR] : 0.1364
##
##                   Kappa : 0.2377
##
##  Mcnemar's Test P-Value : 0.1002
##
##             Sensitivity : 0.4419
##             Specificity : 0.7869
##          Pos Pred Value : 0.5937
##          Neg Pred Value : 0.6667
##              Prevalence : 0.4135
##          Detection Rate : 0.1827
##    Detection Prevalence : 0.3077
##       Balanced Accuracy : 0.6144
##
##        'Positive' Class : 0
##
```

The reduced model, including only Lag1 and Lag2, yields a test error rate comparable to or lower than that of the full model. This suggests that the excluded variables do not significantly improve predictive performance. Hence, the reduced model is preferred due to its parsimony, as it achieves similar accuracy with fewer predictors.